

13.1 Vector Functions and Space Curves

1 Warm Up

Question 1: What is *domain*? and What is the domain of $\ln(x-2)$? $\sqrt{x^2-4}$? $\sqrt{x^2+4}$?

Problem 1: Find $\lim_{x \rightarrow 0} \frac{1-e^x}{x}$

2 Motivation

In two-dimension, The tools of calculus are used on functions and curves. Functions are the language of calculus and it is seemingly natural to try to develop similar functions in 3-dimensions. It turns out that having an output of a function being a vector has many interesting consequences.

3 Definitions & Theorems

1. In 3-D a **parametric function** is of the form $x = f(t), y = g(t), z = h(t)$. Where t is our only independent input (parameter) and for each t in our domain, we get a point. Likewise a **vector function** $r(t)$ takes in our parameter t and outputs a vector where each component is a function.

$$r(t) = \langle f(t), g(t), h(t) \rangle = f(t)i + g(t)j + h(t)k.$$

Since each vector output is in our domain for t , the terminal points of our vector functions outputs, traces out a curve in 3-space which has some orientation. This is known as a **space curve**.

2. We can imagine now taking the **limit** of a vector function in very natural way.

$$\lim_{t \rightarrow a} r(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle.$$

3. We say a vector function is **continuous** at a if the limit of the vector function as t approaches a is equal to the actual vector function value at a .

$$\lim_{t \rightarrow a} r(t) = r(a).$$

4. The vector function $\langle t, t^2, t^3 \rangle$ gives us the **twisted cubic**. The vector function $\langle (2 + \cos 1.5t) \cos(t), (4 + \sin 20t) \sin t, \cos 20t \rangle$ gives the **trefoil knot**.

4 Examples

Problem 1. Write the component function for the following vector function, find its domain, and find any vector output.

$$r(t) = \left\langle \frac{1}{t}, \sqrt{t+2}, \ln(t-3) \right\rangle.$$

Problem 2. Find the $\lim_{t \rightarrow 0} r(t)$ where $r(t) = (1 + t^5)i + te^{-t}j + \frac{\sin t}{t}k$.

Problem 3. *What does the vector function $r(t) = \cos t i + \sin t j + tk$ look like?*

Problem 4. *Find a vector equation and parametric equations for the line segment which joins the point $P(1, 3, -2)$ and $Q(2, -1, 3)$.*

5 Scratch Work